A spiking ring attractor with intermittent (On/Off) decoding

Spiking-network counterpart of the rate model of On/Off states (Panichello et al. 2024 data). Code, parameters and notes: spiking/ on branch claude/gallant-volta-bkjoq (network.py, readout.py, run.py, PARAMETERS.md, NOTE.md).

Summary

- A ring of 2,048 excitatory and 512 inhibitory LIF neurons (Compte et al. 2000) with NMDA-only recurrent excitation (τ_N = 150 ms) holds the cued location as a bump. A common input to all E cells briefly drops ("attention dips"); each dip silences the bump, and it restarts at the cued place when the input returns, from the residual NMDA gating plus a brief overshoot of the input ("kick").
- Read out like the recordings (330 cells per session, L2 logistic decoder, λ = 10,000, shuffle null, the same On/Off labelling), the network gives about 3 Off states per trial with median lengths of 98–122 ms, close to the data (102–129 ms) in all 7 decoded populations, and an Off-length histogram of the data's shape. Without dips there are no Off states for the strong decoders.
- Not matched: decoders are much too strong (μ_{on} 5.7–7 vs 1.1–3.9), so On time is too high and On states too long; top-unit decoders are worse, not better, than random ones; memory is held in 83% of trials (target ≥ 90%), with about 10% restarts at the wrong place; and memory falls with time Off far more steeply than the monkeys' performance. The main cause is that model cells are almost silent outside their preferred location (below 1 Hz), unlike PFC units.

Methods

Network.

- LIF neurons: C = 0.5 / 0.2 nF, g_L = 25 / 20 nS (E / I), $E_L = -70$, $V_{th} = -50$, $V_{reset} = -60$ mV, refractory 2 / 1 ms. External Poisson AMPA drive, 1.8 kHz, 3.1 / 2.38 nS.
- Synapses: AMPA (2 ms, external only), NMDA (rise 2 ms, α = 0.5 ms^{-1}, decay **150 ms**, Mg^{2+} block), $GABA_A$ (10 ms). E→E tuned in angle (Gaussian, $\sigma = 18°$, $J^+ = 3.0$), all other connections uniform. g_{EE} = 0.33 nS (retuned for τ_N = 150 ms; published 0.381 nS at 100 ms), g_{EI} 0.292, g_{IE} 1.336, g_{II} 1.024 nS.
- Heterogeneity (fixed, stratified along the ring so the bump is not pinned): thresholds (SD 0.3 mV), and a fraction $h_i \sim U(0, 1)$ of each E cell's recurrent input through a broad kernel $1 + 0.25 \cos \Delta\theta$ instead of the Gaussian footprint (narrowly to broadly tuned cells).
- Common input m(t) scales the external rate of all E cells: dips with Poisson onsets (7 per s), gamma lengths (mean 150 ms, shape 8.7), depth 50%, 10 ms ramps, none before the cue. Kick: $m_{eff} = m + A(m - \bar{m})$, $\bar{m}$ follows m down instantly and recovers with 150 ms, A = 1.
- Task: 500 ms before the cue, 50 ms cue (current to E cells around one of 8 locations), 1.5 s delay, then a 300 ms "response probe" with the input at 1. Batched LIF integration in numba, dt = 0.1 ms; the E→E sum uses the exact Fourier series of the ring kernel.
- Published values (Compte et al. 2000; Wang 1999) were written from memory because the journal hosts are blocked here; they are marked for verification in PARAMETERS.md.

Read-out (as the data).

- 12 sessions × 480 trials (60 per cue), 330 recorded cells per session (264 E, 66 I) across the ring. Spike counts in 100 ms windows every 10 ms.
- For each cue vs the opposite cue: L2 logistic regression, λ = 10,000, intercept as a penalised feature, trained on correct trials, leave-one-out (exact for the second-order expansion of the loss, which matches full Newton fits to ~1e-4 at this λ); null from 30 label shuffles; decoder output for the cued location z-scored against the null at each time point.
- On = runs of $z > 1.96$ whose summed z exceeds the null's 95th percentile; Off = ≥ 5 points of $z < 0.84$; states in 0–1,400 ms, fractions in 500–1,400 ms.
- Populations: all cells; top 5/20/50 ranked by split-half ANOVA between the two cues of each pair; random 5/20/50 (10 draws).
- Memory held: bump location from the whole E population in the last 100 ms of the delay (spike rate if the bump is active, else NMDA gating), within 22.5° of the cue.

Tuning: what was changed to reach this regime

Starting point: the published ring (Compte et al. 2000; Wang 1999) and, for the common input, the rate model's fitted dips and kick. With the published values and τ_N = 100 ms the network reproduces the published behaviour (spontaneous state ~3 Hz, bump ~50 Hz, slow drift), which served as an implementation check. Each change below was made for the reason given, in this order (tests on 8–32 trials per setting, then 240-trial sessions).

Parameter	Start	Final	Why
NMDA decay τ_N	100 ms	150 ms	Design requirement.

E→E NMDA g_{EE}	0.381 nS	0.33 nS	With τ_N = 150 ms the published value gives a bump before the cue. Scanned for a stable spontaneous state plus a cue-triggered bump; final window 0.30–0.33 nS.
Footprint peak J^+	1.62	3.0	At 1.62 the bistable window was only 0.36–0.375 nS. Raised to 2, then to 3 to compensate the weaker tuned input once the heterogeneity below was added.
Tuning heterogeneity h_i, κ	none	U(0, 1), κ = 0.25	Every cell was sharply tuned, so 5 random cells decoded as well as the top 5. A fraction h_i of each E cell's input goes through a broad kernel 1 + κ cos Δθ. κ ≥ 0.5 made the spontaneous state unstable.
Threshold SD	0	0.3 mV	Heterogeneity required. 1 mV (random draws) seeded spontaneous bumps at clumps of low-threshold cells.
Stratified draws	–	blocks of 16 cells	Random draws pinned the bump ~15° from the cue; each block of 16 neighbouring cells now holds the same quantiles in random order.
Cue	250 ms, weaker	50 ms, 500 pA	Task cue is 50 ms; 200 pA for 50 ms did not reliably start a bump.
Kick A, τ_{kick}	0.8, 15 ms	1.0, 150 ms	A sharp kick makes every E cell fire at once; one spike nearly saturates its NMDA gating, the cued cells lose their advantage and global inhibition silences the ring (memory lost, A ≥ 1). A slow kick restarts the bump; without any kick it fails after 200–250 ms dips.
Dip depth	0.60	0.50	Both silence the bump; 0.6 lost the memory more often (59–81% vs 84–94%).
Dip rate, mean length	10/s, 90 ms	7/s, 150 ms	With 90 ms dips the decoded z rarely stays below 0.84 for 50 ms (100 ms windows bridge the silence): too few Off states. Longer, sparser dips give the data's Off counts and lengths; longer still (5/s × 190 ms) costs memory (77%).
Response probe	–	300 ms at m = 1	Added to check whether a silent bump at the end of the delay can still restart at the cue.

Tried and not kept: stronger external drive and larger threshold spread, to raise E baselines to PFC-like rates; neither raised them above ~1 Hz before bumps formed spontaneously.

Results

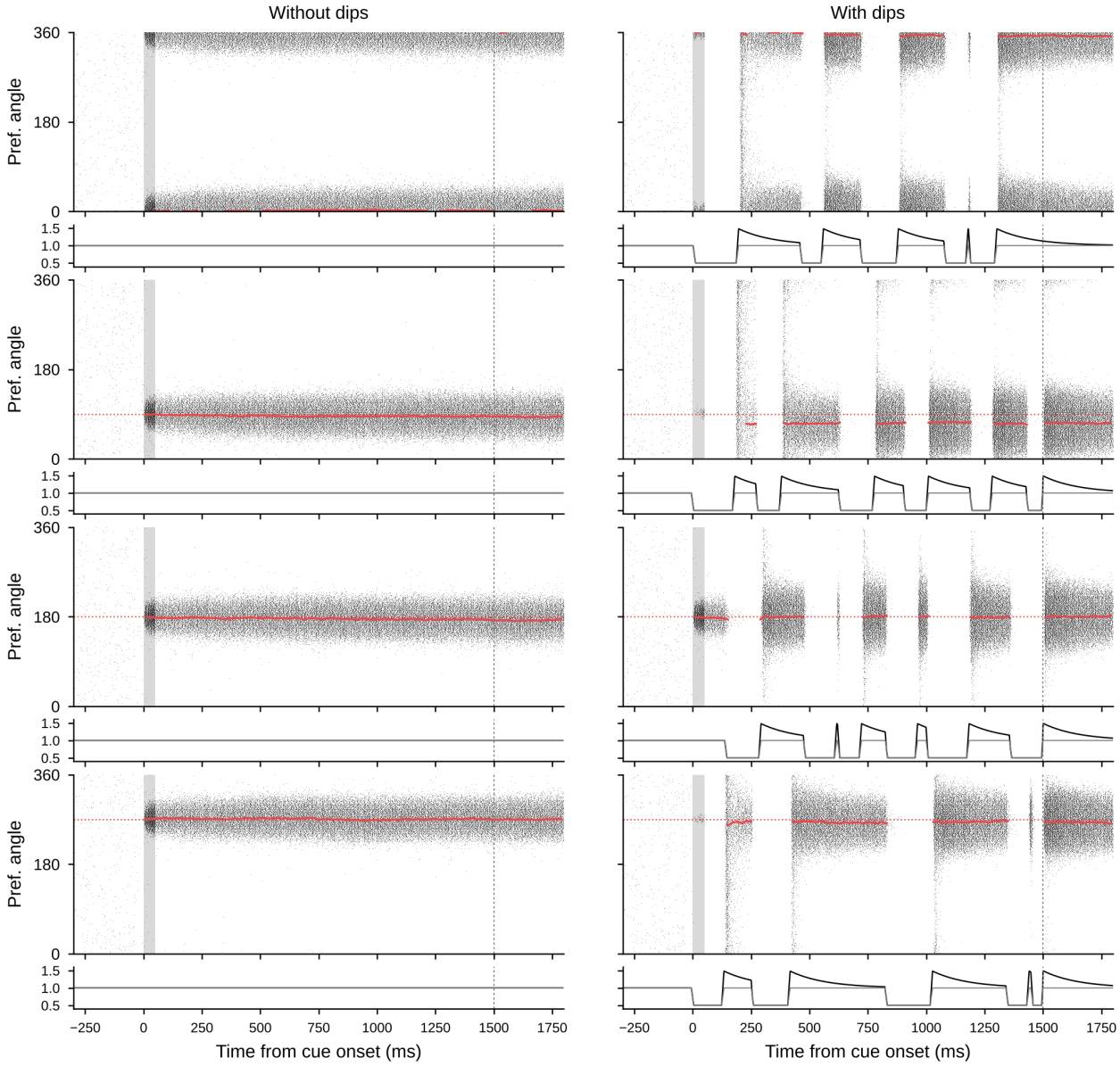


Figure 1. Example trials at four cue locations (0°, 90°, 180°, 270°), without (left) and with (right) dips. Each pair has the same neuron noise; only the common input differs. Raster: E cells by preferred angle; red line: bump position (population vector, 20 ms bins, shown when the bump is active); red dotted line: cue; grey bar: cue; dashed line: end of the delay, followed by the response probe. Below each raster: common input m(t) (grey) and the drive to E cells with the kick (black). Without dips the bump holds the cued place for the whole delay. With dips it is silenced in every dip and restarts at the same place; dips may start at cue onset, in which case the bump first appears at the next recovery. In the 90° trial it restarts about 18° from the cue and stays there (within the 22.5° criterion).

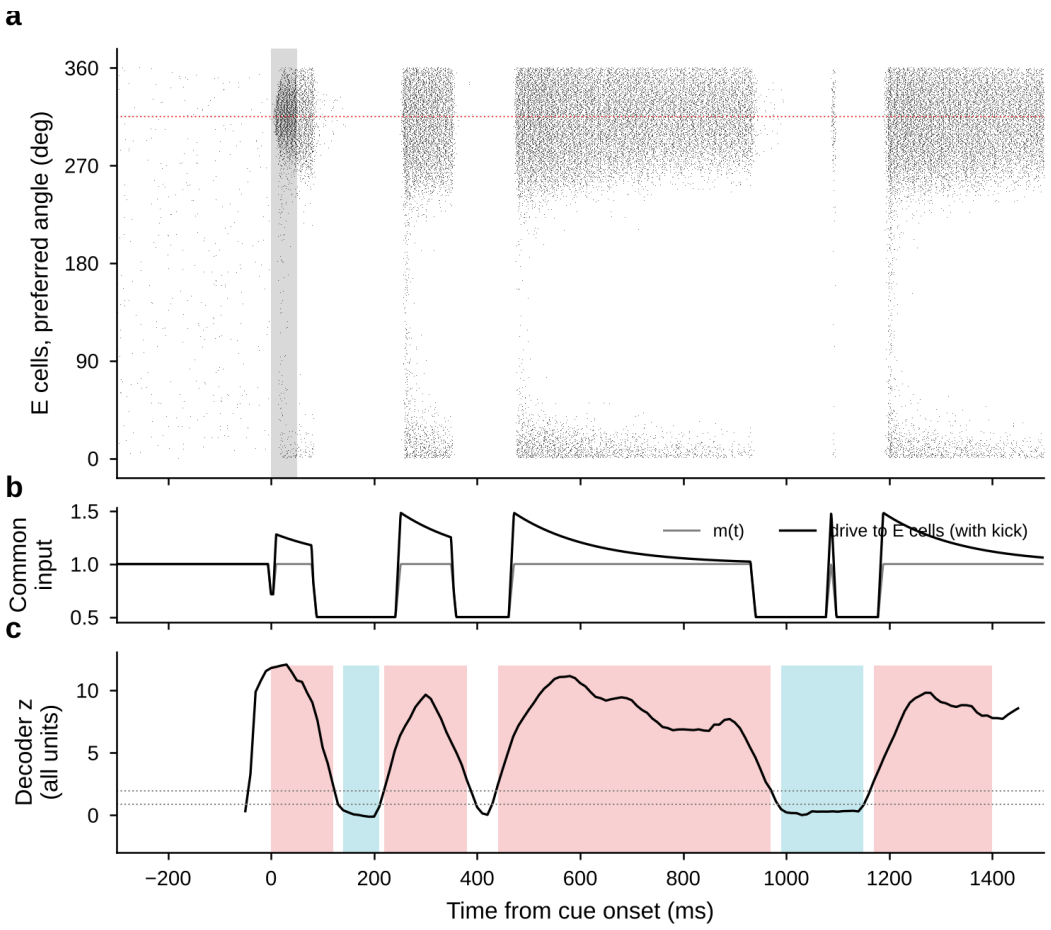


Figure 2. Example trial (one trial in detail, with its decoding). a, raster of the E population sorted by preferred angle (red dotted line: cue). b, common input m(t) and the drive to E cells with the kick. c, all-units decoder z, On (red) and Off (blue) states. Each dip silences the whole bump and z falls to chance; the bump restarts at the cued place when the input returns.

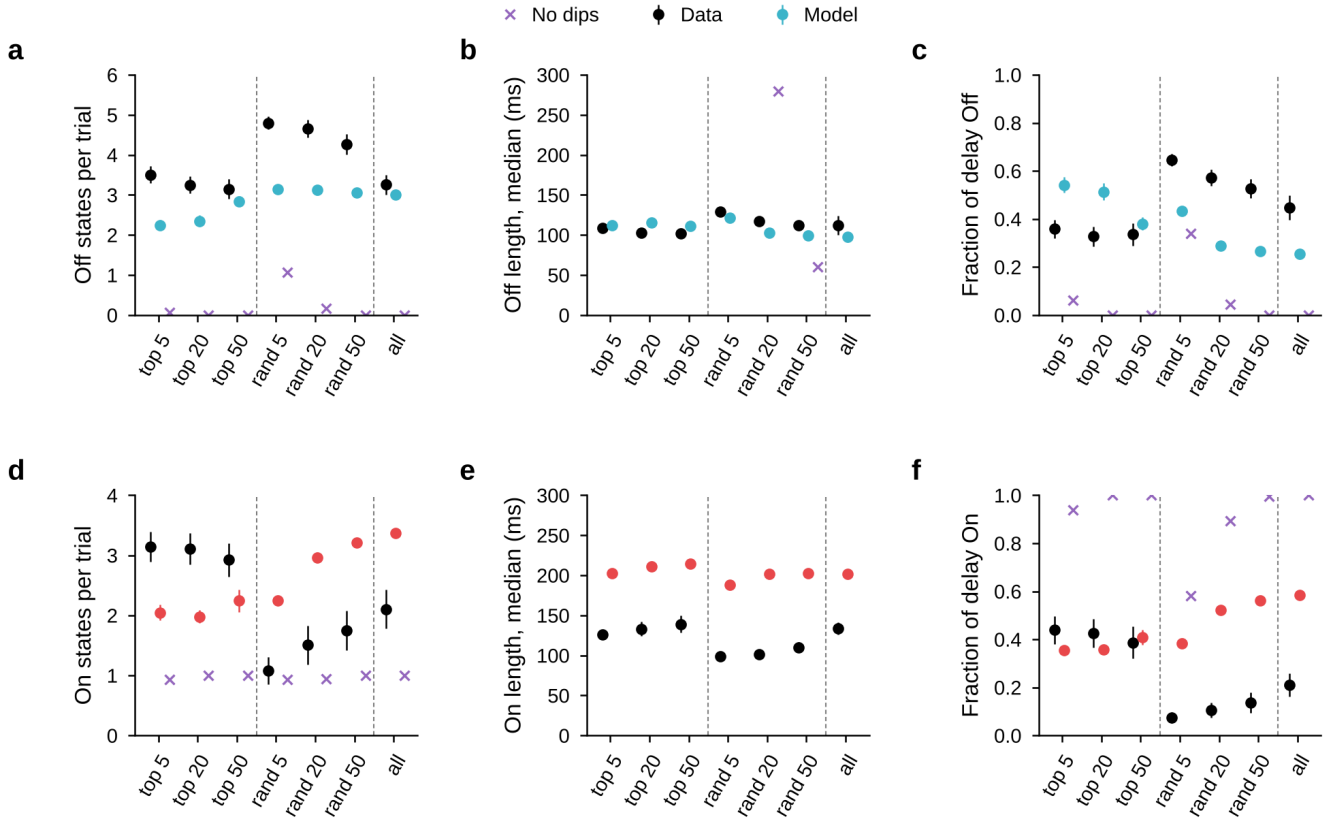


Figure 3. On/Off statistics of the 7 decoded populations. Black: data (mean ± 2 SEM, 25 sessions); colour: model (12 sessions); purple crosses: the same network and read-out without dips. Off counts and lengths are close to the data; the model has too much On time and too little Off time for the strong decoders, and top units are worse than random ones (reverse of the data). Without dips, the strong decoders have no Off states.

Table 1. Model (data in brackets), mean over sessions; lengths in ms.

population	Off/trial	Off median	frac Off	On/trial	On median	frac On	μ_{on}
top 5	2.25 (3.50)	112 (109)	0.54 (0.36)	2.04 (3.14)	202 (126)	0.35 (0.44)	7.00 (3.94)
top 20	2.35 (3.25)	116 (103)	0.51 (0.33)	1.98 (3.11)	211 (133)	0.36 (0.43)	7.12 (3.85)
top 50	2.84 (3.15)	111 (102)	0.38 (0.34)	2.24 (2.92)	215 (139)	0.41 (0.39)	6.11 (3.48)
rand 5	3.15 (4.80)	122 (129)	0.43 (0.65)	2.25 (1.07)	188 (99)	0.38 (0.07)	5.80 (1.14)
rand 20	3.13 (4.65)	103 (117)	0.29 (0.57)	2.97 (1.51)	202 (102)	0.52 (0.11)	5.86 (1.42)
rand 50	3.06 (4.27)	100 (112)	0.26 (0.53)	3.21 (1.75)	203 (110)	0.56 (0.14)	5.78 (1.69)
all	3.00 (3.25)	98 (112)	0.26 (0.45)	3.37 (2.10)	202 (134)	0.59 (0.21)	5.67 (2.18)

Memory held at the end of the delay: **83.2%** (sessions 79.6–87.1%); bump restarts at the cue in the response probe: 88.7%; bump at the wrong place: 10.1%.

Table 2. No-dip control (6 sessions × 480 trials), same read-out. Memory 100%.

population	Off/trial	Off median	frac Off	On/trial	On median	frac On	μ_{on}
top 5	0.06	1410	0.06	0.94	1410	0.94	12.66
top 20	0.00	–	0.00	1.00	1410	1.00	11.52
top 50	0.00	–	0.00	1.00	1410	1.00	11.30
rand 5	1.07	518	0.34	0.94	1040	0.58	12.46
rand 20	0.17	280	0.04	0.95	1410	0.89	13.13
rand 50	0.00	60	0.00	1.00	1410	1.00	12.76
all	0.00	–	0.00	1.00	1410	1.00	10.85

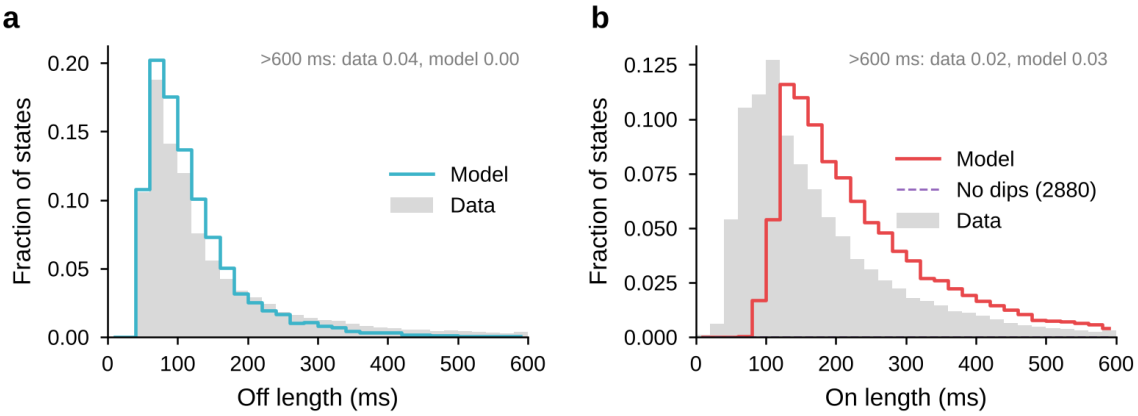


Figure 4. State lengths, all units (20 ms bins, pooled). The Off histogram matches the data (rise at 40–60 ms, peak at 60–80 ms, long tail). On states are about 70 ms too long and almost never shorter than 100 ms.

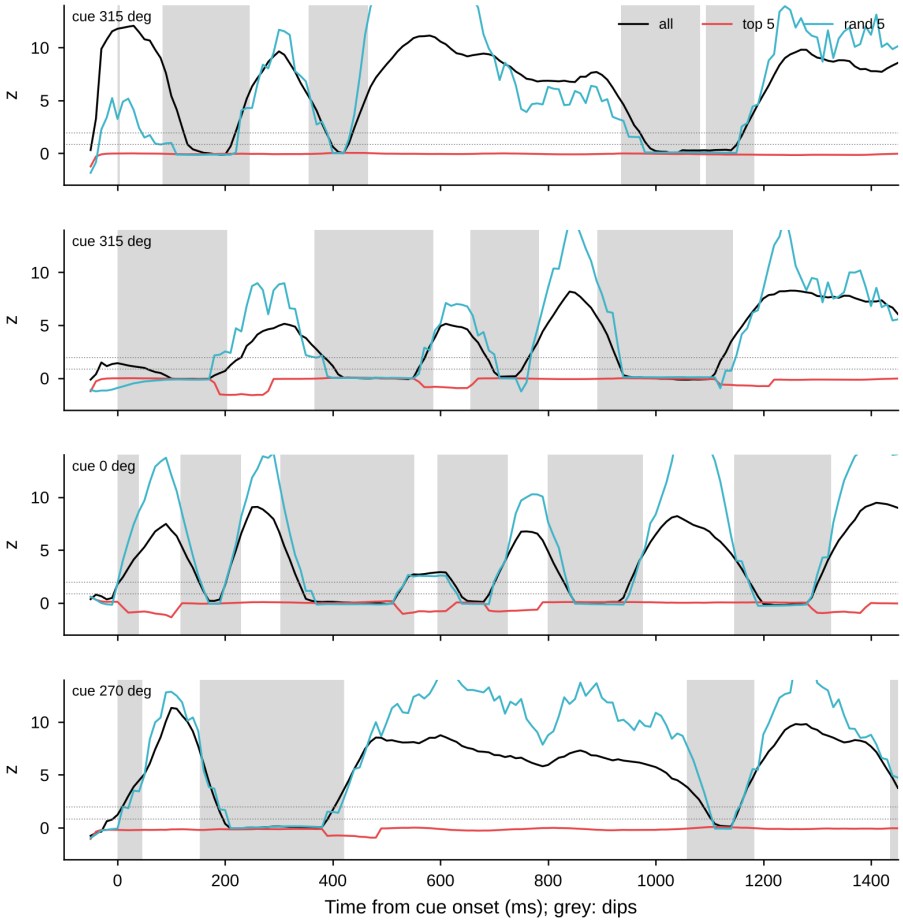


Figure 5. Decoder z of three populations on the same trials (grey: dips). All units and random 5 cells follow the bump; the top 5 cells of this pair prefer the other cue and stay silent, so z ≈ 0 and the trial is Off throughout.

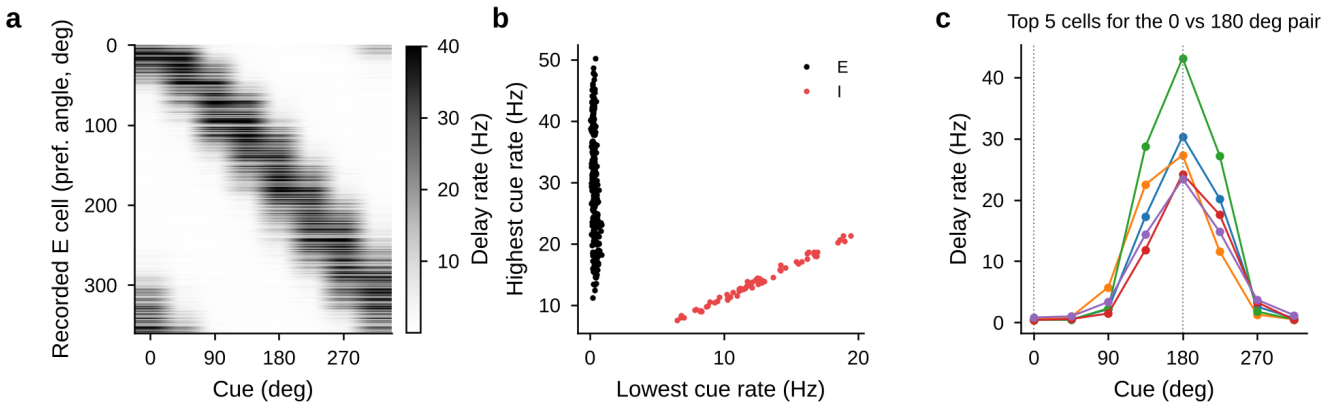


Figure 6. Why top units are not better. a, delay rate of the recorded E cells against cue: narrow tuning. b, highest vs lowest cue rate: E cells are almost silent at their non-preferred cues (below 1 Hz); interneurons are weakly modulated. c, the top 5 cells for the 0° vs 180° pair all prefer 180°. Without dips their silence identifies the other cue; with dips the preferred cue's trials are often silent too, so silence carries no information. In PFC, cells have baselines of several Hz, so suppression stays informative.

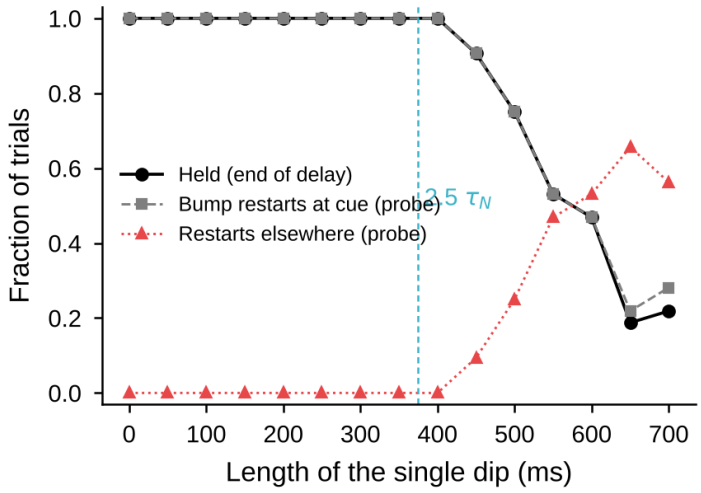


Figure 7. Memory after a single dip (32 trials per length, dip 600 ms after the cue, no other dips). The memory survives dips up to 400 ms and is lost by about 575 ms (50%), later than 2.5 τ_N = 375 ms because the slow kick also bridges long silences. Lost memories mostly restart at the wrong place.

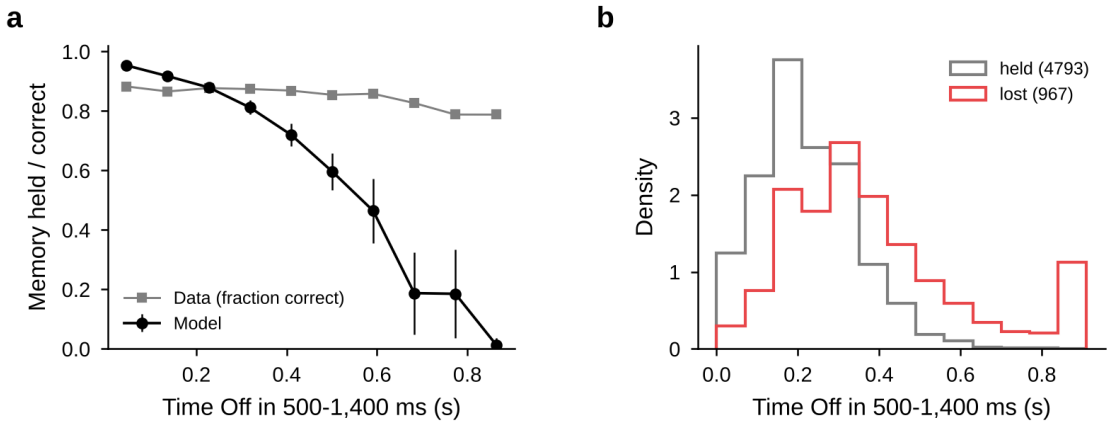


Figure 8. Memory held against time Off in the delay (all-units decoder trained on all trials, 5,760 trials). a, model (black) and the monkeys' fraction correct (grey). b, distribution of time Off for held and lost trials. More time Off goes with memory loss, as in the data, but far more steeply.

Checks

- Stability (no dips, 2 s before the cue): no spontaneous bump (E 0.6 Hz, I 4.5 Hz), bump peak 58 Hz, drift about 1° per s, 15/16 held.

- Brian2 cross-check (same equations, explicit all-to-all synapses, reduced 512 E + 128 I ring): E rate 16.4 vs 18.1 Hz, I 9.9 vs 9.8 Hz, bump peak 108 vs 112 Hz (numba vs Brian2): agreement within about 10%.

Main differences from the rate model and the data

- A sharp kick (15 ms, as in the rate model) makes every E cell fire at once; one spike nearly saturates the NMDA gating, which erases the memory. The kick had to be slow (150 ms) and the dips longer and sparser (7 per s, 150 ms, depth 0.5, vs 10 per s, 90 ms, 0.6) for Off states to last ≥ 50 ms in 100 ms windows.
- Model cells are far more selective than PFC units and have almost no baseline; raising baselines (stronger drive, more threshold spread) made the bump form before the cue. This explains the too-strong decoders and the inverted top vs random order. A more strongly driven ring, with baselines of several Hz, would be the next step.
- Memory 83% (target ≥ 90%), with wrong-place restarts (~10%), which the rate model never shows; more Off time costs more memory than in the rate model.
- Open questions: whether the data ranked top units per cue pair or over all 8 cues; parameter values to be checked against the PDFs of Compte et al. (2000) and Wang (1999).